

MUNGLINUP WEST, Australia

WMO: 956440

Lat: 33.555S

Lon: 120.700E

Elev: 723

StdP: 14.32

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.7	41.1	27.1	21.6	66.2	30.6	25.5	65.2	27.7	58.8	25.4	57.8	8.2	320	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.3	96.4	65.9	91.8	64.8	87.3	64.5	71.4	83.5	69.0	81.4	67.3	79.2	12.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	105.8	75.6	65.7	97.5	71.6	64.1	92.1	69.6	35.7	82.9	33.6	81.8	32.2	79.0	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	22.0	19.9	36.3	106.4	1.4	3.4	35.3	108.8	34.5	110.8	33.8	112.7	32.8	115.1		
(5)			35.0	74.1	1.3	3.6	34.1	76.7	33.3	78.9	32.6	80.9	31.6	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	69.6	69.8	67.5	63.8	58.2	54.2	52.4	53.5	55.7	60.1	64.1	66.8	(6)
(7)		DBStd	8.28	6.25	5.85	6.02	5.76	4.50	3.70	3.70	4.57	5.45	6.27	6.65	6.47	(7)
(8)		HDD50	45	0	0	0	0	0	5	17	14	8	1	0	0	(8)
(9)		HDD65	2065	17	16	38	90	218	325	391	358	284	180	97	51	(9)
(10)		CDD50	4156	608	554	544	415	254	131	91	122	180	315	422	520	(10)
(11)		CDD65	699	160	150	117	54	6	0	0	1	6	29	69	107	(11)
(12)		CDH74	7144	1628	1346	1007	430	56	3	1	32	114	438	837	1252	(12)
(13)		CDH80	3152	805	608	427	125	6	0	0	5	31	167	363	615	(13)
(14)	Wind	WSAvg	10.7	11.9	11.8	10.5	9.8	9.7	10.2	10.3	10.5	10.6	11.5	11.4	(14)	
(15)	Precipitation	PrecAvg	19.6	1.1	1.3	1.1	1.3	2.0	2.0	2.5	2.4	2.1	1.8	1.5	1.1	(15)
(16)		PrecMax	25.9	7.1	4.5	4.3	3.3	5.1	5.2	5.7	6.3	4.6	3.9	4.3	3.3	(16)
(17)		PrecMin	11.5	0.0	0.0	0.2	0.3	0.4	1.0	0.8	0.6	0.3	0.3	0.1	0.0	(17)
(18)		PrecStd	3.6	1.5	1.2	1.0	0.7	1.2	1.0	1.2	1.2	1.1	1.0	0.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.1	99.9	96.9	89.0	80.0	72.3	70.5	78.1	84.5	92.1	96.3	101.8	(19)
(20)			MCWB	66.0	68.9	65.5	62.5	58.7	56.1	54.6	56.0	60.2	61.5	64.1	64.2	(20)
(21)		2%	DB	95.7	93.2	90.4	83.8	74.5	68.1	66.8	71.1	76.7	84.5	89.4	93.4	(21)
(22)			MCWB	66.7	66.4	64.6	61.9	57.5	55.0	53.5	54.8	58.0	60.1	62.3	64.8	(22)
(23)		5%	DB	88.7	87.8	84.3	78.8	70.6	65.0	63.5	66.5	71.0	78.2	83.5	86.9	(23)
(24)			MCWB	67.2	66.9	64.1	61.3	56.7	53.9	53.1	54.0	56.1	59.9	62.7	63.9	(24)
(25)		10%	DB	82.6	82.2	78.9	74.6	67.3	62.1	60.7	63.1	66.8	72.9	77.7	80.5	(25)
(26)			MCWB	66.1	65.5	62.9	60.4	56.3	53.3	52.0	53.4	55.1	58.7	60.7	62.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.5	74.5	70.5	67.5	63.6	60.1	58.4	59.8	64.4	68.2	71.8	73.9	(27)
(28)			MCDWB	87.4	85.1	82.0	76.9	70.2	66.5	64.4	67.6	75.1	80.0	83.4	86.5	(28)
(29)		2%	WB	71.4	70.3	68.4	65.4	61.3	58.0	56.2	57.6	60.3	64.3	66.9	68.3	(29)
(30)			MCDWB	83.3	85.1	79.5	73.8	67.5	63.5	61.9	65.5	70.6	75.0	80.1	81.5	(30)
(31)		5%	WB	68.6	68.1	66.7	63.8	59.4	56.4	54.6	55.8	58.0	61.7	64.4	65.7	(31)
(32)			MCDWB	80.9	80.8	76.4	71.8	66.0	61.4	60.5	63.2	66.7	72.4	76.7	79.6	(32)
(33)		10%	WB	66.5	66.5	65.0	62.3	57.7	54.8	53.1	54.2	56.2	59.7	62.3	63.7	(33)
(34)			MCDWB	78.3	78.2	73.9	70.3	64.4	59.9	58.7	61.1	64.8	70.2	73.9	76.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.4	22.3	21.2	20.2	17.7	16.1	16.1	18.1	20.9	22.7	23.3	24.2	(35)
(36)			MCDBR	35.3	32.3	29.7	26.4	23.0	20.3	20.8	24.6	28.6	32.5	34.3	36.5	(36)
(37)			MCWBR	11.4	10.6	9.5	10.1	10.4	10.4	11.1	12.2	12.4	13.3	12.6	12.2	(37)
(38)		5% WB	MCDBR	27.8	26.2	23.7	20.5	17.7	16.6	17.4	20.2	24.3	26.9	28.1	29.4	(38)
(39)			MCWBR	12.3	10.5	9.6	9.6	10.2	10.4	10.8	11.6	12.5	13.9	13.5	13.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.342	0.324	0.313	0.298	0.289	0.286	0.295	0.315	0.323	0.330	0.333	(40)	
(41)		taud	2.543	2.566	2.605	2.628	2.636	2.634	2.644	2.608	2.532	2.536	2.540	2.556	(41)	
(42)		Ebn at Noon	314	309	304	291	279	274	281	293	302	312	318	320	(42)	
(43)		Edh at Noon	35	33	29	26	23	21	22	25	31	33	35	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2349	2057	1702	1271	953	809	876	1152	1539	1927	2209	2431	(44)	
(45)		RadStd	158	141	110	77	60	32	43	46	58	116	139	156	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (8 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-108	+167	+79	

Nomenclature: See separate page